Experiment No: 4

Date:

QR CODE

Experiment Title

Analysis, Design, and Implementation of Smart Safety Companion Using Mobile Sensors

Aim

To analyze, design, and implement a Smart Safety Companion Android application that uses built-in smartphone sensors to automatically detect abnormal motion and generate emergency alerts with location and direction details.

Introduction

Smart Safety Companion is an Android-based prototype application developed using MIT App Inventor. The main objective of this application is to improve personal safety by using smartphone sensors. The application continuously monitors device movement, location, and orientation. When abnormal motion is detected, the system automatically generates an emergency alert. This project demonstrates the practical application of sensor-based mobile systems for real-time safety monitoring.

Problem Description

In many emergency situations such as accidents, sudden falls, or health issues, a person may not be able to manually call for help. Delay in communication and absence of accurate location information can increase the risk and reduce response efficiency. Existing safety systems often require user interaction or additional hardware. Hence, there is a need for an automatic, mobile-based safety system that works independently using built-in sensors.

Objectives

1. To monitor user movement using accelerometer sensor.
2. To detect abnormal or sudden motion automatically.
3. To capture real-time location and device direction.
4. To generate emergency alerts without user intervention.
5. To design a simple and user-friendly interface.

Proposed Solution

The proposed system continuously observes accelerometer values to calculate shake intensity. When the shake value exceeds a predefined threshold, the system assumes an emergency situation. It then captures GPS location, device direction, and current time. An alert message is generated and displayed. A cooldown mechanism is implemented to avoid repeated alerts.

Development Environment

<u>Software Platform:</u>

MIT App Inventor is used to design the user interface and backend logic using block programming.

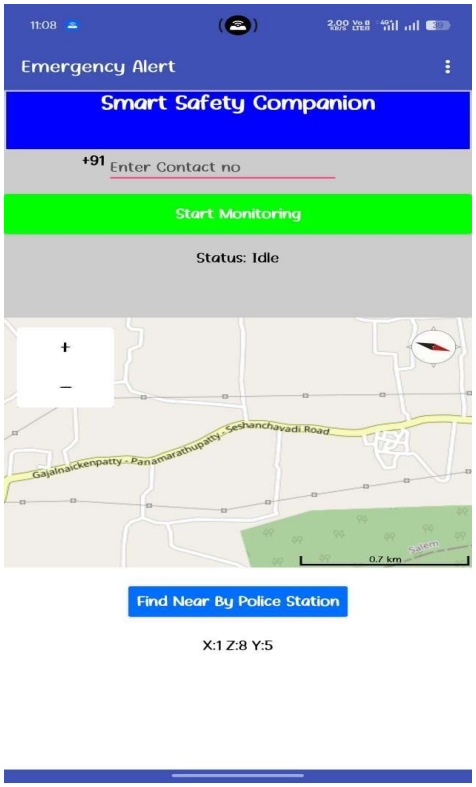
Hardware Platform:
Android smartphone with built-in sensors.

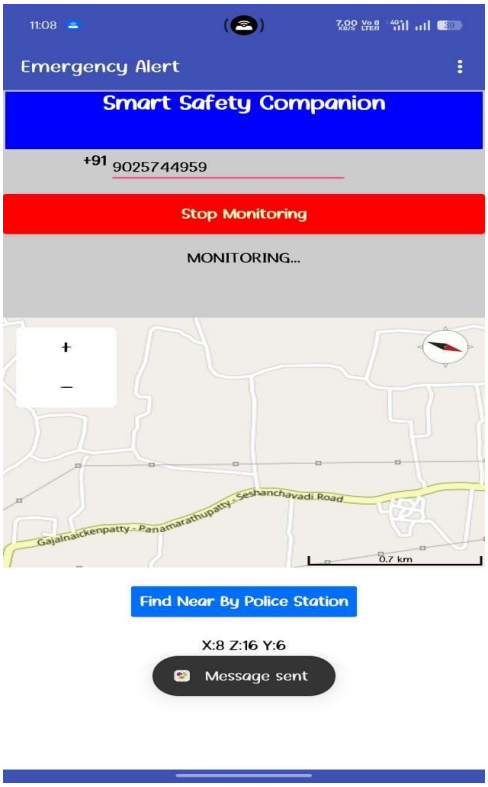
Sensors Used:
- Accelerometer Sensor
- Location Sensor
- Orientation Sensor

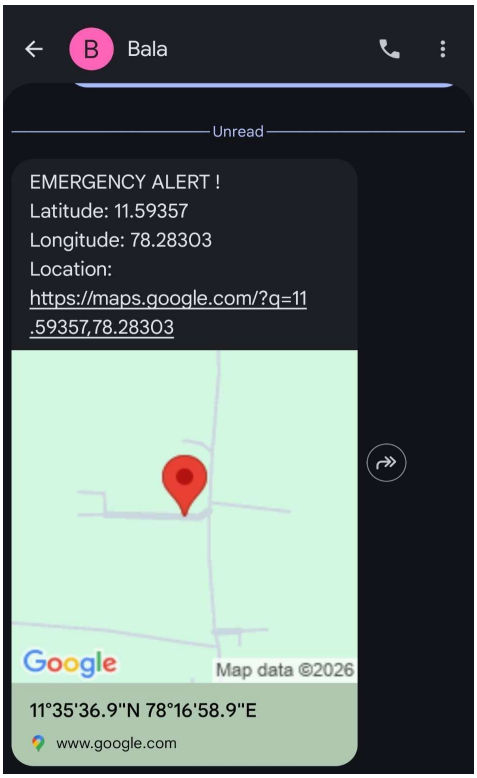
Components Used:
- Buttons
- Labels
- TextBox
- Clock

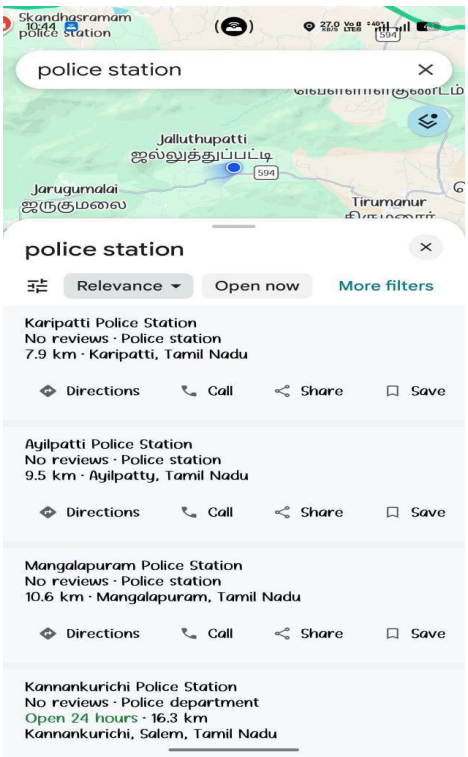
Application Interface

The application interface is designed to be simple and easy to understand. It contains a TextBox to enter an emergency contact number. The Start Monitoring button activates all sensors, and the Stop Monitoring button deactivates them. Labels are used to display sensor values, system status, and emergency alerts.









Working Methodology

When the Start button is pressed, accelerometer, location, and orientation sensors are enabled. The accelerometer provides X, Y, and Z axis values continuously. The shake intensity is calculated using the formula:

Shake Value = |X| + |Y| + |Z|

If the shake value exceeds the threshold, abnormal motion is detected. The system then captures latitude, longitude, device direction, and time. An emergency alert is generated. A cooldown delay prevents repeated alerts.

Backend Implementation

The backend logic is implemented using MIT App Inventor blocks.

<u>Sensor Control:</u>
The Start button enables all sensors, while the Stop button disables them.

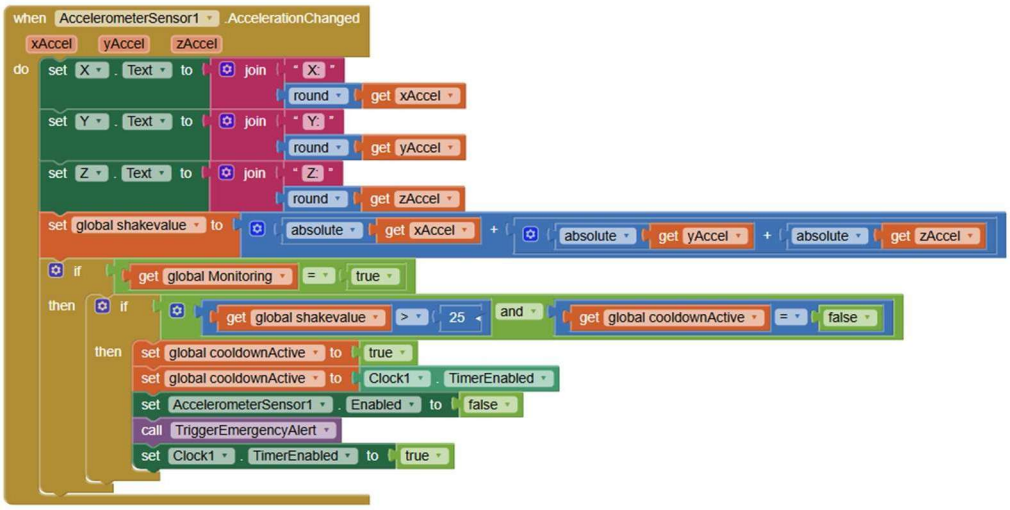
<u>Motion Detection</u>:
Accelerometer values are processed to calculate shake intensity and compared with a threshold.

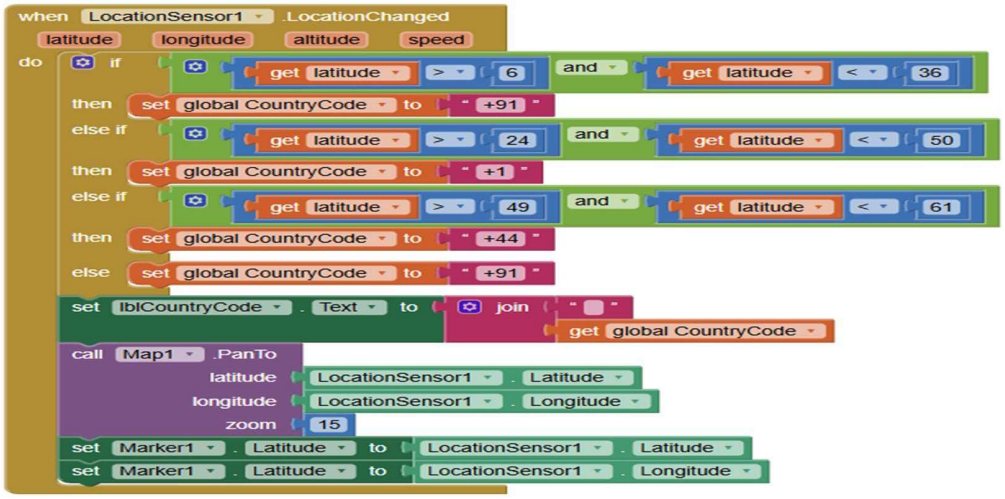
Alert Generation:

Once emergency is detected, an alert message is generated using collected data.

Cooldown Control:

The Clock component ensures alerts are not repeated frequently.





Observation

- Sensor values change dynamically with device movement.
- Sudden shakes produce higher shake values.
- Location is captured accurately.
- Alerts are generated only when threshold is crossed.
- Cooldown mechanism works effectively.

Advantages

- Automatic emergency detection
- No external hardware required

- Easy to use
- Real-time location tracking
- Improves safety response time

Limitations

- Works only on Android devices
- Requires GPS and sensor permissions
- Accuracy depends on sensor quality
- Internet may be required for location services

Future Enhancements

- Automatic SMS sending to emergency contacts
- Integration with police and ambulance services
- Voice-based emergency alert
- Cloud-based alert storage
- AI-based motion analysis

Result

The Smart Safety Companion application was successfully designed and implemented. The system accurately detects abnormal motion and generates emergency alerts with location details.

Conclusion

This experiment demonstrates the effective use of smartphone sensors for personal safety applications. The Smart Safety Companion provides an automated and reliable safety solution. The project highlights the capability of MIT App Inventor in developing real-world Android applications.